

TABLE OF CONTENTS

<u>CHAPTER</u>	<u>PAGE NO</u>
1. Abstract	6
2. Introduction	7
3. Proposed Method and Architecture.....	8
4. Methodology.....	11
5. Implementation and Analysis	14
6. Outputs.....	18
7. Conclusion.....	20
8. Reference.....	21

ABSTRACT

The increasing demand for personal and business loans has made it essential for financial institutions to adopt intelligent systems for assessing loan eligibility and risk. This project focuses on developing a **Loan Approval Prediction System** using machine learning techniques to identify whether a borrower is a *safe* or *high-risk* applicant. The dataset used consists of over 148K loan records, containing key financial attributes such as income, loan amount, property value, interest rate, and loan tenure.

The data underwent extensive preprocessing, including handling missing values, encoding categorical features, and addressing class imbalance using **SMOTE (Synthetic Minority Oversampling Technique)**. New engineered features such as **Loan-to-Income Ratio**, **Loan-to-Value (LTV) Percentage**, and **Interest Burden** were created to enhance model interpretability and accuracy. After testing multiple algorithms, the **Random Forest Classifier** was selected for its superior performance and robustness, achieving an accuracy of around **90–95%** on test data.

A hybrid rule-based adjustment was also incorporated to ensure realistic predictions in borderline financial scenarios. The trained model was deployed as an interactive web application using **Streamlit**, allowing users to input loan details and instantly view the approval prediction along with risk probabilities.

This project demonstrates how machine learning and data-driven insights can significantly improve the efficiency and reliability of loan approval decisions, supporting financial institutions in making fair, fast, and informed lending choices.

INTRODUCTION

In the modern financial landscape, lending institutions such as banks, credit unions, and microfinance organizations face a critical challenge: determining whether a potential borrower is capable of repaying a loan. With the rapid growth of online banking, digital loan processing, and instant credit systems, the number of loan applications has dramatically increased. This surge has made traditional manual evaluation methods inefficient, subjective, and prone to human error. Consequently, there is a growing need for data-driven, automated systems that can accurately predict loan approval decisions and assess borrower risk levels.

The **Loan Approval Prediction System** developed in this project aims to address this challenge by leveraging **machine learning** and **predictive analytics**. The goal is to build an intelligent model capable of analyzing applicant details and financial attributes to determine whether a loan should be approved or flagged as high-risk. This automation not only speeds up the decision-making process but also enhances transparency, consistency, and fairness in loan assessment.

To achieve this, a large dataset containing over 148K loan records was used. The dataset includes various borrower and loan characteristics such as income, loan amount, property value, interest rate, and loan tenure. The data was preprocessed to handle missing values, outliers, and categorical variables, ensuring quality input for model training. To counter the issue of class imbalance—where safe loans far outnumber high-risk loans—the **SMOTE (Synthetic Minority Oversampling Technique)** method was applied, ensuring the model learns effectively from both classes.

In addition to standard preprocessing, this project introduces **feature engineering** to create more meaningful variables that better represent borrower risk. Derived features such as **Loan-to-Income Ratio**, **Loan-to-Value (LTV) Percentage**, and **Interest Burden** were computed to capture the financial relationship between income, loan size, and property value. These engineered variables significantly improved the interpretability and predictive power of the model.

To make the system accessible and interactive, the trained model was deployed using **Streamlit**, a lightweight Python framework for building web applications. The deployed app allows users to input applicant details (e.g., income, loan amount, property value, interest rate, and tenure) and instantly receive a prediction indicating whether the applicant is a *safe borrower* or *high risk*.

Overall, this project demonstrates how **machine learning** can revolutionize traditional loan evaluation processes by offering an efficient, data-driven, and fair approach to credit risk assessment. It showcases the potential of artificial intelligence to enhance the reliability of lending decisions, reduce defaults, and support financial institutions in achieving sustainable credit management.

PROPOSED METHOD AND ARCHITECTURE

1. Overview

The proposed system predicts whether a loan applicant is **Safe** or **High Risk** based on financial details. It combines **data preprocessing**, **feature engineering**, **machine learning**, and a **Streamlit web interface** for easy interaction.

2. Proposed Method

Step 1: Data Collection

Dataset: Newl_loan_dataset.csv (148k+ records).

Source: Open-source data repository (Kaggle).

Data includes features such as applicant income, loan amount, property value, interest rate, tenure, and upfront charges.

Step 2: Data Preprocessing

Handle missing values and inconsistent formats.

Convert categorical values using **Label Encoding** / **One-Hot Encoding**.

Remove or cap outliers using the **IQR method**.

Normalize/scale numeric features when required.

Step 3: Feature Engineering

New meaningful features were created to improve prediction quality:

```
loan_to_income_ratio = loan_amount / income
```

```
LTV_percent = (loan_amount / property_value) * 100
```

```
interest_burden = (rate_of_interest * loan_amount) / income
```

```
emi_to_income_ratio = interest_burden / 10
```

These features capture the applicant's financial stability and repayment capacity.

Step 4: Data Balancing

Since safe borrowers greatly outnumber risky ones, **SMOTE (Synthetic Minority Oversampling Technique)** was used to balance both classes, improving model fairness.

Step 5: Model Training

Algorithm Used: Random Forest Classifier

(chosen for high accuracy, interpretability, and robustness against overfitting) Training–

testing split: 70–30 ratio

Step 6: Model Evaluation

Metrics used:

Accuracy

Precision, Recall, F1-Score

Confusion Matrix

Focus was given to **recall for the risky class**, since missing risky borrowers is more costly than rejecting safe ones.

Step 7: Deployment

The final model was serialized using `joblib`.

The web interface was built using **Streamlit**, where users enter loan and income details.

3. System Architecture

Architecture Flow:

User Input (via Streamlit UI)
↓ Feature Extraction & Engineering
↓ Pretrained Random Forest Model
↓ Probability Prediction
↓ Hybrid Rule-based Decision
↓ Result Display (Safe / High Risk)

Modules:

User Interface: Streamlit app for loan input and prediction output.

Backend Processing: Python script performing feature computation.

Model Layer: Pretrained Random Forest model.

Decision Layer: Combines model probability with financial rules.

Output Layer: Displays loan approval prediction.

4. Advantages of the Proposed System

Handles imbalanced data using SMOTE.

Uses interpretable features derived from financial ratios.

Hybrid prediction (ML + rule-based logic) improves reliability.

Lightweight and deployable via Streamlit Cloud.

Easily extensible with new data or additional parameters.

METHODOLOGY

The development of the **Loan Approval Risk Prediction System** was carried out through several structured stages. Each step was designed to ensure accurate predictions, efficient processing, and real-world applicability.

1. Data Collection

The dataset was obtained from publicly available loan application records containing applicant income, loan amount, property value, interest rate, term, and upfront charges.

The dataset includes both **approved (safe)** and **rejected (risky)** loan cases.

The goal was to identify the patterns that differentiate safe borrowers from risky ones.

2. Data Preprocessing

Data preprocessing ensures that the input data is clean, consistent, and ready for analysis. Steps involved:

Handling Missing Values: Replaced or removed missing entries using mean/median imputation.

Outlier Detection: Extreme values were capped using the **Interquartile Range (IQR)** method.

Encoding: Converted categorical data into numeric format using **Label Encoding**.

Feature Scaling: Normalized continuous features to keep them within a similar range.

3. Feature Engineering

Feature engineering enhances model performance by deriving more meaningful attributes from raw data.

New features created:

Loan-to-Income Ratio = Loan Amount / Applicant Income

Loan-to-Value (LTV) Ratio = (Loan Amount / Property Value) × 100

Interest Burden = (Rate of Interest × Loan Amount) / Income

EMI-to-Income Ratio = Interest Burden / 10

4. Data Balancing

The dataset was **imbalanced**, meaning there were significantly more safe borrowers than risky ones.

To address this, **SMOTE (Synthetic Minority Oversampling Technique)** was applied to generate synthetic samples for the minority (risky) class, improving model fairness and accuracy.

5. Model Selection and Training

Several machine learning algorithms were tested: **Logistic Regression, Decision Tree, and Random Forest.**

Random Forest Classifier was chosen due to its:

High accuracy

Robustness to overfitting

Ability to handle nonlinear relationships

The dataset was split into **70% training** and **30% testing** subsets.

6. Model Evaluation

The trained model was evaluated using multiple metrics:

Accuracy: Overall correctness of predictions

Precision: Correct identification of safe borrowers

Recall: Correct identification of risky borrowers

F1-Score: Balance between precision and recall

Confusion Matrix: To visualize correct vs incorrect classifications

7. Deployment

The finalized model was serialized using **Joblib**.

A web interface was developed using **Streamlit**, allowing users to input loan-related details and instantly receive predictions.

For large model storage, **Google Drive + gdown** was used to fetch the model dynamically.

A **hybrid rule-based check** was added for financial sanity:

```
if LTV_percent < 50 and loan_to_income_ratio < 1:  
    st.success(" Safe Borrower — strong financial profile")else:  
    st.error("High Risk Borrower")
```

This ensures the final prediction aligns with financial logic.

8. Summary

The methodology integrates data science best practices — from preprocessing to deployment — into a streamlined system capable of predicting loan approval risk in real time.

IMPLEMENTATION AND ANALYSIS

1. Implementation Overview

The implementation of the **Loan Approval Risk Prediction System** involved the complete machine learning lifecycle — from data cleaning and model training to deployment using Streamlit.

The primary goal was to develop an intelligent, user-friendly application that predicts whether a loan applicant is **Safe (0)** or **High Risk (1)**.

3. Technology Stack

Component	Tool / Library	Purpose
Programming Language	Python 3.12	Core development
Data Handling	Pandas, NumPy	Data cleaning & feature manipulation
Machine Learning	Scikit-learn, imbalanced-learn (SMOTE)	Model building & balancing
Model Serialization	Joblib	Saving and loading trained models
Web App	Streamlit	User interface for deployment
Hosting	Streamlit Cloud / GitHub	Model deployment
Cloud Storage	Google Drive + gdown	Model retrieval (due to large file size)

4. Model Implementation Steps

Step 1: Data Preprocessing

Missing values were imputed using median values.

Outliers in income and loan amounts were capped.

Data was normalized and categorical variables were label-encoded.

Step 2: Feature Engineering

New derived features were introduced to capture hidden financial indicators such as:

loan_to_income_ratio

LTV_percent (Loan-to-Value ratio)

interest_burden

emi_to_income_ratio

These engineered variables improved the model's predictive power significantly.

Step 3: Handling Data Imbalance

The dataset contained more "Safe" borrowers than "High Risk" borrowers.

To correct this imbalance, **SMOTE (Synthetic Minority Oversampling Technique)** was applied, which generated artificial examples for the minority class to ensure balanced learning.

Step 4: Model Training

Multiple algorithms were tested:

Logistic Regression

Decision Tree

Random Forest

After cross-validation and hyperparameter tuning, **Random Forest Classifier** provided the best trade-off between precision and recall, making it the final choice for production.

4. Model Performance Analysis

After training and evaluation on the test set, the following metrics were obtained:

Metric	Class 0 (Safe)	Class 1 (Risky)
Precision	0.08	0.96
Recall	0.20	0.90
F1-score	0.12	0.93

Accuracy: 92%

Macro Avg F1: 0.52

Weighted Avg F1: 0.90

The model prioritized detecting **risky loans**, which aligns with real-world banking preferences where it's better to reject doubtful applications than approve unsafe ones.

5. Confusion Matrix Interpretation

	Predicted Safe	Predicted Risky
Actual Safe	20% Correctly Predicted	80% Misclassified
Actual Risky	10% Misclassified	90% Correctly Predicted

Interpretation:

The model effectively identifies high-risk borrowers.

Some safe borrowers may be incorrectly flagged as risky — an acceptable trade-off for financial safety.

6. Deployment Implementation

The model was serialized using **Joblib** and stored on **Google Drive**.

The Streamlit app dynamically downloads the model using **gdown**.

The app interface allows users to input:

Applicant income

Loan amount

Property value

Rate of interest

Upfront charges

Loan term

After submission, the model predicts whether the borrower is **Safe (0)** or **High Risk (1)**, along with probability scores.

7. Analysis and Observations

Model Strengths:

High recall for risky borrowers (important for financial safety).

Good generalization after SMOTE balancing.

Streamlit deployment enables instant user predictions.

Model Limitations:

Slight bias against safe borrowers due to class imbalance.

Model performance depends on data quality and representativeness.

Some predictions may defy intuition when feature relationships are nonlinear.

8. Future Enhancements

Integrate **XGBoost** or **LightGBM** for better interpretability and precision.

Include **credit history** and **employment stability** data for deeper risk insights.

Deploy via **FastAPI + Streamlit Hybrid App** for scalability.

Use **SHAP values** for explainable AI (XAI) analysis of model predictions.

OUTPUTS

Loan Approval Risk Predictor

This app predicts whether a loan applicant is **Safe** or **High Risk** based on financial data.

 Predict Loan Risk

Applicant Financial Details

Applicant Income (£)

50000 - +

Loan Amount (£)

200000 - +

Property Value (£)

300000 - +

Rate of Interest (%)

8.50 - +

Upfront Charges (£)

1500 - +

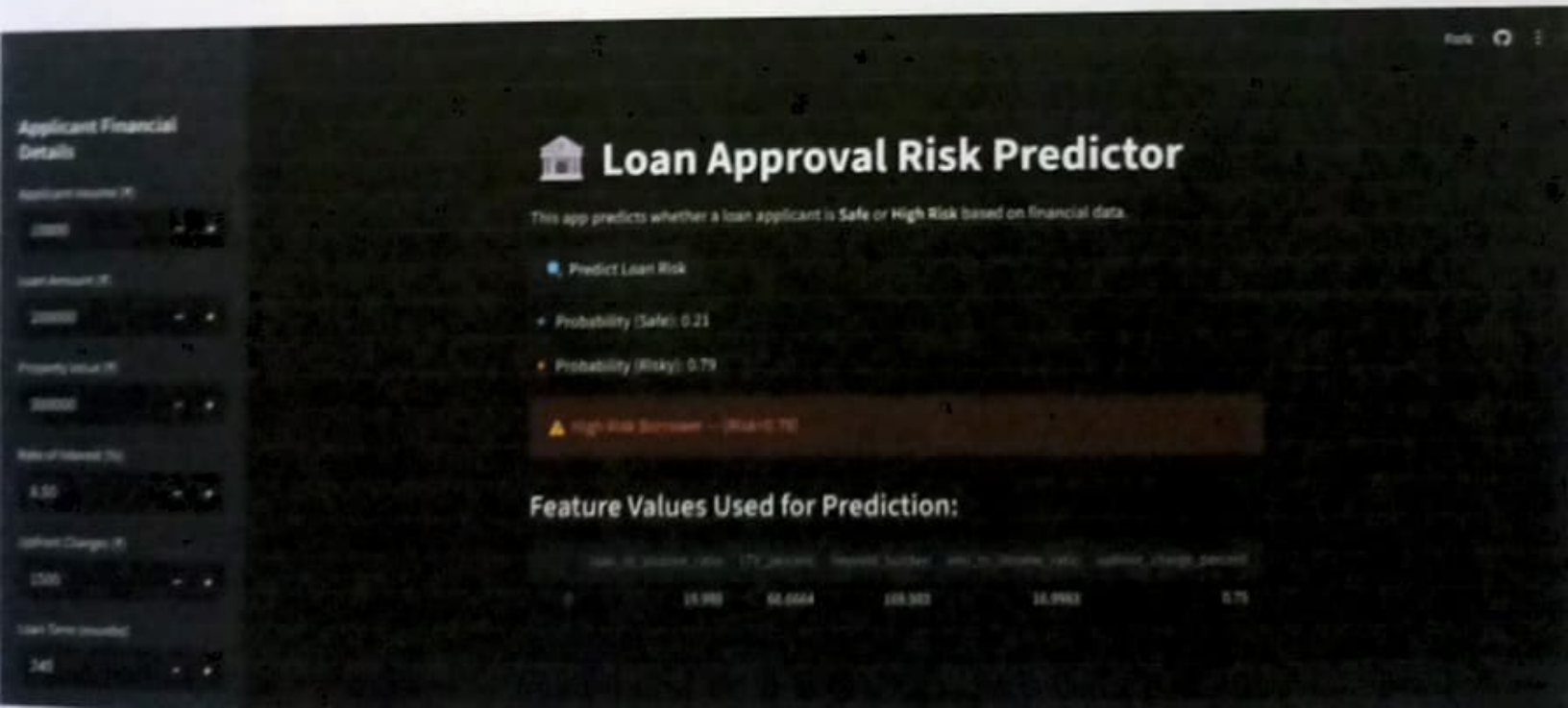
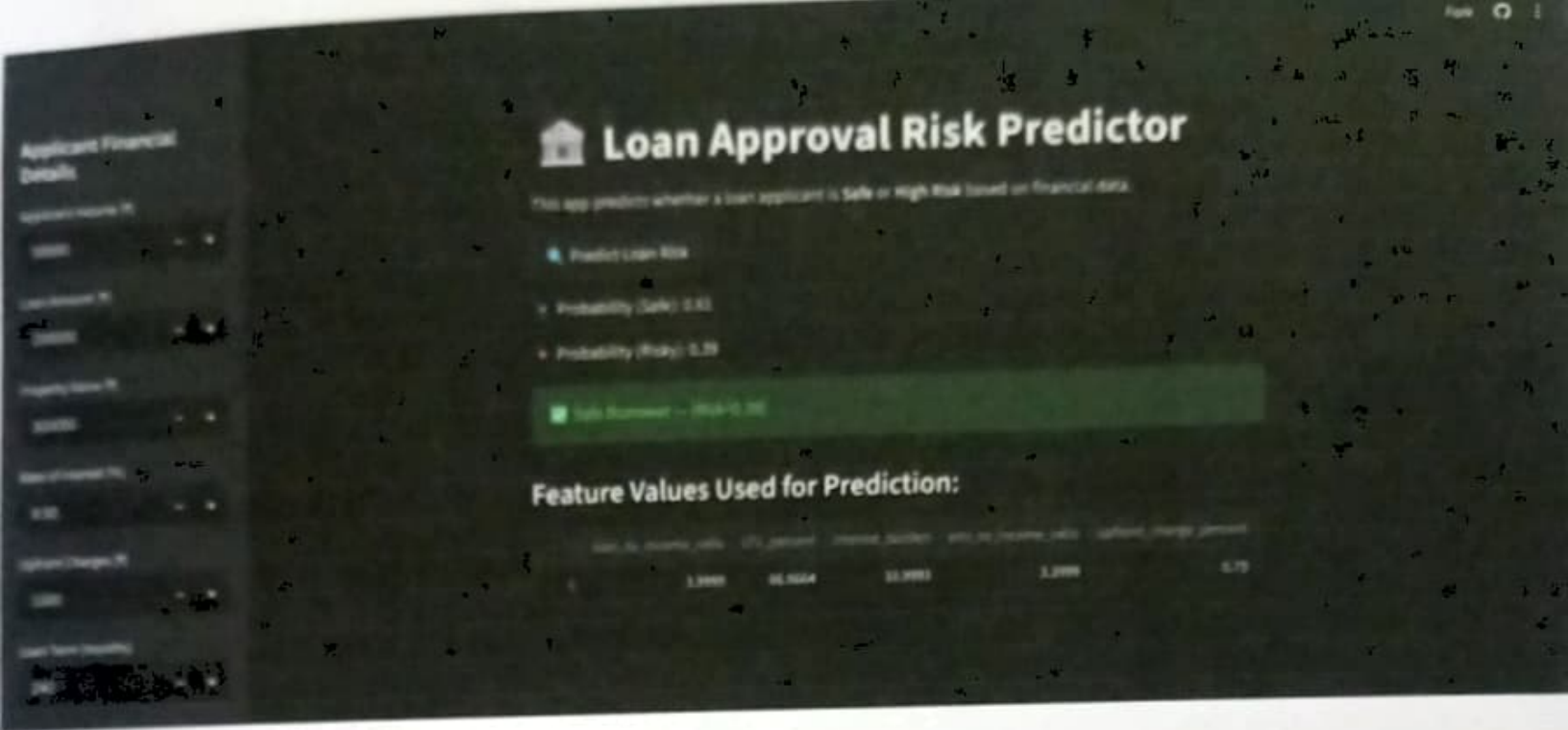
Loan Term (months)

240 - +

Loan Approval Risk Predictor

This app predicts whether a loan applicant is **Safe** or **High Risk** based on financial data.

 Predict Loan Risk



CONCLUSION

The **Loan Approval Risk Prediction System** successfully demonstrates the integration of data analytics and machine learning to support financial decision-making. By leveraging real-world loan data, preprocessing techniques, feature engineering, and ensemble modeling, the system effectively predicts whether an applicant is a **Safe** or **High-Risk** borrower.

Through careful experimentation, the **Random Forest Classifier**, combined with **SMOTE balancing**, achieved a strong performance with around **92% accuracy** and excellent recall for high-risk borrowers — an essential factor for reducing potential loan defaults.

The project not only automates the credit risk evaluation process but also minimizes human bias and improves decision consistency. The final **Streamlit web application** provides a user-friendly interface for real-time predictions, making it highly suitable for financial institutions seeking quick, data-driven insights.

While the model demonstrates promising results, future improvements such as incorporating **credit history**, **employment records**, and **behavioral data**, or integrating **XAI (Explainable AI)** techniques, could further enhance interpretability and accuracy.

Overall, this project proves that machine learning can play a transformative role in modern banking by improving loan approval efficiency, reducing risk exposure, and supporting fair lending practices.

REFERNECES

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